

Yunxi Lai

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Current affiliation: Zhongshan Ophthalmic Center, Sun Yat-sen University, China

EDUCATION

Ophthalmology | Zhongshan Ophthalmic Center, Sun Yat-sen University, China

Jul 2023–Jun 2026 **Doctor of Medicine (M.D.)** | Degree conferral expected December 2026

Thesis: Transformation from Clinical Data to Scientific Evidence Enabled by Large Language Models in Ophthalmology.

Sep 2020–Jun 2023 **Master of Medicine (M.M.)** | Licensed Physician

Standardized Residency Training Certificate in Ophthalmology

Thesis: Analyses of Routine Preoperative Blood Testing in Ocular Patients.

Clinical Medicine | Guangdong Medical University, China

Sep 2015–Jun 2020 **Bachelor of Medicine**

RESEARCH EXPERIENCE

Generative AI

Development, Validation and Implementation of an Ophthalmic Large Language Model

Jul 2023–Jun 2026

- **Defined ophthalmic evaluation benchmarks** for adapting a general-purpose LLM to ophthalmology and assessing its question-answering performance; co-authored an article published in *JAMA Ophthalmology* in 2024.
- **Conceived its clinical application as a pre-consultation tool**, designed an agent to actively elicit and document medical histories, and integrated it into existing outpatient workflows and hospital systems.
- **Led full validation in a randomized controlled trial**, including protocol and statistical analysis plan, ethics approval and trial registration, Python coding, statistical analysis in R and other trial operations; co-first-authored an article accepted by *npj Digital Medicine* on 30 August 2026.
- **Coordinated multidisciplinary collaboration and dissemination** across grant applications, intellectual property and public outreach, contributing to app deployment serving over 1.7 million people across four countries and to the project's selection as a Chinese digital health case study at the 2025 World Health Assembly.

Epidemiology

Routine Preoperative Blood Testing in Ophthalmic Surgery

Sep 2021–Jun 2023

- **Independently conducted a retrospective study** evaluating the necessity of routine preoperative blood testing and the applicability of *Choosing Wisely* recommendations to ophthalmic surgery in China.
- **Analyzed and interpreted data from 102,542 patients**, resulting in a first-author publication in the *British Journal of Ophthalmology* in 2024.

Cohort Studies	Pediatric Cohort on Ocular Effects of Systemic Medications
Aug 2023–Mar 2026	- Coordinated a cross-institutional ophthalmology–pediatrics collaboration to monitor ocular adverse effects associated with systemic medications, spanning ethics approval, participant recruitment, ophthalmic examinations, and longitudinal data management. - Co-established and managed a prospective cohort of 600+ children, with an average of seven follow-up visits per participant. - Designed an LLM-assisted extraction method with active data-quality control to address inefficient longitudinal data extraction, converting discharge summaries into structured research variables; internally evaluated using retrospective longitudinal inpatient records.
Nov 2023–Mar 2025	Community-Based Ophthalmic Cohorts - Explained test results, offered medical advice and collaborated with local healthcare providers.

TRANSFERABLE SKILLS

Clinical Visual Assessment: Experienced in comprehensive ophthalmic examination and imaging, including slit-lamp biomicroscopy, ophthalmoscopy, optical coherence tomography and scanning laser ophthalmoscopy; fine visuomotor coordination and procedural precision; demonstrated adaptability and commitment to high-quality work through surgical training and rotations in emergency, inpatient and outpatient settings.

Research Design: Designed and conducted human-subject studies, including retrospective analyses, randomized controlled trials and prospective cohorts, from protocol development and registration to statistical analysis planning, data collection and analysis.

Quantitative and Computational Skills: Python and R programming for data processing, quantitative analysis and computational research; SPSS for statistical analysis; AI-assisted coding and prototyping using Codex and Visual Studio Code.

Project Management and Interdisciplinary Collaboration: Coordinated timelines and deliverables across clinical, technical and industry teams; aligned responsibilities and expectations.

Research Presentation: Drafted, revised and published two first-author manuscripts in peer-reviewed journals; gave and received constructive scientific feedback.

Languages: Chinese (native), English (fluent).

TEACHING & OUTREACH

Instructor	Pre-Rotation Training for First-Year Ophthalmology Residents
Aug 2024	- Prepared teaching materials and taught essential clinical procedures.
Contributor	Community Engagement & Outreach
2020-2025	- Led visual identity and art direction for the official launch of the ophthalmic LLM; contributed to video planning, content development and media promotion. - Planned and contributed to community activities, including Physicians' Day events, cultural performances and public eye-health screening initiatives.

CONFERENCE PRESENTATION

May 2025 **Evaluating GPT-4o in ophthalmic consultation and documentation: Bridging knowledge with clinical practice.** (Poster) *Association for Research in Vision and Ophthalmology (ARVO) Annual Meeting, Salt Lake City, Utah, USA.*

PUBLICATIONS

Peer-reviewed Journal Articles: 7; First-author: 2

1. Luo, M. J.*, **Lai, Y.***, Wang, W.*, Bi, S., Jin, L., Chen, J., Wu, X., Lin, Z., Wang, J., Pang, J., Dongye, M., Wu, Y., Liu, Z., & Lin, H. (Accepted August 30, 2026; *equal contribution). Standardized Pre-consultation by a Large Language Model Agent vs Ophthalmology Residents: A Randomized Clinical Trial. *npj Digital Medicine*.
2. **Lai, Y.**, Zeng, W., Liao, J., Yu, Y., Liu, X., & Wu, K. (2024). Retrospective Analyses of Routine Preoperative Blood Testing in a Tertiary Eye Hospital: Could Choosing Wisely Work in China? *British Journal of Ophthalmology*, 108(6), 897–902.
3. Luo, M. J., Pang, J., Bi, S., **Lai, Y.**, Zhao, J., Shang, Y., Cui, T., Yang, Y., Lin, Z., Zhao, L., Wu, X., Lin, D., Chen, J., & Lin, H. (2024). Development and Evaluation of a Retrieval-Augmented Large Language Model Framework for Ophthalmology. *JAMA Ophthalmology*, 142(9), 798–805.
4. Luo, M. J., Bi, S., Pang, J., Liu, L., Tsui, C. K., **Lai, Y.**, Chen, W., Yang, Y., Xu, K., Zhao, L., Jin, L., Lin, D., Wu, X., Chen, J., Chen, R., Liu, Z., Zou, Y., Yang, Y., Li, Y., & Lin, H. (2025). A Large Language Model Digital Patient System Enhances Ophthalmology History Taking Skills. *npj Digital Medicine*, 8(1), 502.
5. Liu, L., Hong, J., Wu, Y., Liu, S., Wang, K., Li, M., Zhao, L., Liu, Z., Li, L., Cui, T., Tsui, C. K., Xu, F., Hu, W., Yun, D., Chen, X., Shang, Y., Bi, S., Wei, X., **Lai, Y.**, Lin, D., Fu, Z., Deng, Y., Cai, K., Xie, Y., Cao, Z., Wang, D., Zhang, X., Dongye, M., Lin, H., & Wu, X. (2024). Digital ray: Enhancing Cataractous Fundus Images Using Style Transfer Generative Adversarial Networks to Improve Retinopathy Detection. *British Journal of Ophthalmology*, 108(10), 1423–1429.
6. Shang, Y., Lin, Y., Li, R., Shang, Y., Li, M., Zhao, L., Jin, L., Xu, A., Liu, D., Wu, Q., Luo, M., Pang, J., Bi, S., He, Y., Xu, M., Chen, X., Cao, Z., Yu, S., Zhao, J., **Lai, Y.**, Chen, W., & Lin, H. (2025). The Effectiveness of Large Language Models in Medical AI Research for Physicians: A Randomized Controlled Trial. *Cell Reports Medicine*, 6(12), 102469.
7. Li, Y., Cui, T., Lin, Z., Zhao, L., **Lai, Y.**, Yun, D., Lin, Z., Li, M., Huang, Y., Chen, S., Wang, D., Qiu, Y., Xu, Y., Yang, Y., Lin, D., Jiang, X., & Lin, H. (2026). Risk Factors and Critical Thresholds for Glucocorticoid-Induced Ocular Adverse Reactions in Children: Implications for Muscle and Neuromuscular Disease Management. *European Journal of Translational Myology*, 36(2).